

Clinical *Graph-JEPA* — a patient-state World Model

We move past extract-and-store. The graph becomes a **world model** of the patient: it represents state, predicts the consequences of clinical actions, and judges its own structural coherence. Three model layers — extraction, consequence reasoning, and graph physics. Data: **MIMIC-IV-Note**.

START	GATE A	GATE B (FREEZE)	PENS-DOWN	SUBMIT
Thu May 28	Mon Jun 1	Fri Jun 5	Wed Jun 10	Fri Jun 12

*A static knowledge graph records what was said. A **world model** represents the patient, simulates what an intervention would do, and knows when its own structure is implausible.*

LeCun JEPA → V-JEPA 2 → action-conditioned latent planning.
EHRWorld · CLARITY (2026): treatment-conditioned patient trajectories in latent space.

THE THREE WORLD-MODEL LAYERS

LAYER 1 · PERCEIVE Extraction → <i>state</i> GLiNER2 · evidence-bound	LAYER 2 · PREDICT Reasoning <i>with consequences</i> action-conditioned latent dynamics	LAYER 3 · REFINE Physics of <i>the graph</i> Graph-JEPA · energy / geometry
GLiNER2 turns the transcript into the patient's observed state: typed entities (symptom, diagnosis, med, lab, history, behavior, anatomy), each pinned to its source quote. This is the encoder of the world model — raw observation → structured latent state. Frontier basis: the "state" half of a JEPA world model — perception that yields a representation a predictor can act on.	The reasoning layer builds relations as consequence-bearing dynamics, not flat edges. Conditioned on a clinical action (start lithium, escalate dose), it predicts the next patient-state subgraph in latent space: if this, then that. Symptom→indicates→diagnosis becomes one slice of a trajectory the model can roll forward. Frontier basis: EHRWorld & CLARITY (2026) — treatment-conditioned, counterfactual patient-state transition in latent space. Value-guided JEPA planning (LeCun et al., 2026).	The graph has a physics: coherent patient-states sit in low-energy regions of latent space; incoherent ones do not. Graph-JEPA predicts masked subgraphs and scores every candidate edge by how well it fits the whole. Geometry is the advantage — hyperbolic embedding encodes clinical hierarchy (infection ⊃ covid-19) so a relation is judged against structure, not just co-occurrence. Frontier basis: Graph-JEPA latent consistency (Skenderi 2023); energy-based structural plausibility; hyperbolic KGE implausibility scoring (2025–26).

MODEL STACK — THE WM LAYERS, NOT AN LLM PIPELINE

L1 · EXTRACTION GLiNER2 schema-driven NER → typed evidence-bound state	L2 · CONSEQUENCE REASONING Action-conditioned latent predictor next-state subgraph given a clinical action	L3 · GRAPH PHYSICS Graph-JEPA + hyperbolic geometry energy / plausibility scoring of the structure
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Scope rule: this is a world-model architecture, not a candidate-LLM bake-off. GLiNER2 is the fixed extractor; the research contribution is Layers 2 and 3 — turning the graph into a predictive, self-scoring patient-state model. **No new baseline-model evaluation.**

WORKSTREAMS — PARALLEL, ONE OWNER EACH

Amit WS1 · DATA	Deliver the MIMIC cohort, the action vocabulary, and the curated ground-truth state-graphs the model is judged against.	FEEDS L1 · L2
May 28-29	Confirm MIMIC-IV-Note access & de-id; select cohort; freeze note set.	
May 30-31	Define typed node/edge schema + the clinical action vocabulary (the interventions Layer 2 conditions on). Start ground-truth state-graph curation.	
Jun 1-3	Finish curated GT slice; sign off state-representation numbers on MIMIC.	
Jun 6-9	Clinical-accuracy review; cut unsupported claims; Schrager framing.	
Kushagra WS2 · TRAINING	Train the consequence-reasoning predictor (L2) and the Graph-JEPA physics layer (L3) .	OWNS L2 · L3
May 28-31	Run GLiNER2 to produce the typed state-graph; confirm format feeding L2/L3.	
Jun 1-3	L2 + L3 v1 : action-conditioned next-state predictor; Graph-JEPA masked-subgraph training with hyperbolic target. First edge-plausibility + consequence scores. DEPENDS ON → GATE A	
Jun 4-5	Wire scores into refinement: re-rank / flag edges; expose latent consequence rollouts. Hand to evaluation. FEEDS → WS4 BY EOD JUN 5	
Jun 6-9	Methods write-up; reproduce checkpoints from clean state.	
Nalin WS3 · DEVOPS	Keep the three-layer pipeline + training loop reproducible and fast ; own the eval harness plumbing.	SPANS L1-L3
May 28-31	Stand up reproducible pipeline on MIMIC: GLiNER2 serving, run/seed logging, artifact capture.	
Jun 1-3	Training infra & checkpoint loop for L2/L3; eval harness wired end-to-end. UNBLOCKS → WS2, WS4	
Jun 4-7	Run scaling, latent-rollout serving, artifact capture so numbers are regenerable.	
Jun 8-9	Reproducibility check : clean checkout reproduces every reported number.	
Summer WS4 · EVALUATION	Own all metrics: state quality, consequence-prediction validity, and structural-coherence gains, plus ablations.	MEASURES L1-L3
May 30-31	Build intrinsic harness: entity/relation F1 + population/relation/schema completeness (state quality, L1).	
Jun 1-3	Stand up L2/L3 evaluation: consequence-prediction (held-out next-state / masked-subgraph accuracy) and coherence (edge-plausibility ranking, retrieval lift). Lock ablations. DEPENDS ON → GATE A	
Jun 4-5	Score refined vs unrefined graphs; with-physics vs without ablation; significance. CONSUMES ← WS2 · DRIVES GATE B	
Jun 6-7	Finalize all tables & figures from frozen results.	
Yining WS5 · PAPER	Own the manuscript end-to-end; frame the world-model contribution and assemble one voice.	FRAMES ALL
May 28-31	Lock outline + the WM positioning (state → consequence → physics); intro, related work (EHRWorld, CLARITY, JEPA), methods skeleton.	
Jun 1-5	Draft methods & setup in parallel with the runs; collect section inputs from WS1–WS4.	
Jun 6-7	Full draft with frozen Gate-B results, tables, figures.	
Jun 8-9	Internal review, Schrager framing, revisions. → PENS-DOWN JUN 10	

GATE A

Mon Jun 1

Layer 1 state-graph reproduces on MIMIC. Entity/relation/completeness agreed. L2/L3 training does not start until the state representation holds.

GATE B

Fri Jun 5 EOD

Results freeze. Whatever consequence + coherence numbers exist by EOD is what the paper reports. After: hardening only, no new claims.

PENS-DOWN

Wed Jun 10

Manuscript + code locked. Jun 11 buffer. Submit Jun 12.